

stopwords, punctuation, and some special words like ‘is’, ‘the’ and ‘of’. Then, to normalise the vocabulary of the corpus, a dictionary-based lemmatization has been used with the Python module spaCy.

```

from gensim import corpora, models
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer
import string
import nltk
import spacy
import matplotlib.pyplot as plt
import pandas as pd
from collections import Counter

nltk.download('punkt_tab')
## False
nlp = spacy.load("en_core_web_sm")

documents = [ "Knowledge of statistics is essential for data
analysis.",

               "The concept of statistics is also helpful for
economics.",

               "The idea of economics is essential for social
study." ,

               "Data analysis is an essential part of machine
learning."
            ]

# Tokenization
texts = [word_tokenize(doc.lower()) for doc in documents]
# Remove punctuation
words_no_punct = [word for lstitm in texts for word in lstitm
if word not in string.punctuation]
# Remove stopwords
process_corpus = []
alist = []
for list_word in texts:
    for listitm in list_word:
        if listitm not in string.punctuation:
            alist.append(listitm)
        process_corpus.append(alist)
    alist=[]
# Remove "is" "the", "of"
process_corpus2 = []
alist = []
for list_word in process_corpus:
    for listitm in list_word:
        if listitm not in ['is', 'the', 'of']:
            alist.append(listitm)
        process_corpus2.append(alist)
    alist=[]
# Remove stopwords
stopwords = set (stopwords.words("english"))
vocabulary = set()
v = []
alist = []
for list_word in process_corpus2:
    for listitm in list_word:
        if listitm not in stopwords:
            alist.append(listitm)
    v.append(alist)
    vocabulary.update(alist)
    alist=[]
# Convert to a sorted list
vocabulary = sorted(list(vocabulary))
print("Vocabulary:", vocabulary)
## Vocabulary: ['also', 'analysis', 'concept', 'data',
'economics', 'economics.the', 'essential', 'helpful',
'idea', 'knowledge', 'learning', 'machine', 'part', 'social',
'statistics', 'study']
filtered_doc = [token.lemma_ for token in nlp("
".join(vocabulary))]
# Create a dictionary and corpus
filtered_doc = [filtered_doc]
dictionary = corpora.Dictionary(filtered_doc)
print("Token to id: ",dictionary.token2id)
## Token to id: {'also': 0, 'analysis': 1, 'concept': 2,
'datum': 3, 'economic': 4, 'economics.the': 5, 'essential':
6, 'helpful': 7, 'idea': 8, 'knowledge': 9, 'learn': 10,
'machine': 11, 'part': 12, 'social': 13, 'statistic': 14,
'study': 15}
corpus = [dictionary.doc2bow(text) for text in texts]
print("Corpus: ",corpus,'\n')
## Corpus: [[(1, 1), (6, 1), (9, 1)], [(0, 1), (2, 1), (5,
1), (6, 1), (7, 1), (8, 1), (13, 1), (15, 1)], [(1, 1), (6,
1), (11, 1), (12, 1)]]
# Print corpus
for i, doc in enumerate(corpus):
    print(f"Document {i+1}: {doc}")
    print('\n')
## Document 1: [(1, 1), (6, 1), (9, 1)]
##
##
## Document 2: [(0, 1), (2, 1), (5, 1), (6, 1), (7, 1), (8,
1), (13, 1), (15, 1)]
##
##
## Document 3: [(1, 1), (6, 1), (11, 1), (12, 1)]
# Prepare frequency table

```